

ChristosHadjinikolis

Engineering Manager | ML Systems Lead | UCL Research Associate

Ph.D. Computer Science | Active Committee Expert @ ISO/CEN-CENELEC JTC'21

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mediums



leadership

people management,

hiring/interviews, mentoring,

code pairing, sprint

reviews/retros, delivery

governance

practice

streaming strategy,

standardisation, **DORA metrics**,

model/data versioning,

observability, auditability

technical

production ML,

streaming/lambda backends,

Java/Kotlin, Python, SQL,

Kafka/Flink, MLflow, Airflow,

AWS/GCP

public work

AI standards (JTC'21);

maritime AI research (UCL);

Agile DS talks; **dynamicio**;

skeleton-replay/IDE plugin

about me

Engineering Manager and technical ML systems leader with 16 years of combined industry and academic experience across AI research, production ML, live data processing, streaming platforms, and backend engineering. Lead a **five-person MLE/DS/DE pod** responsible for most of Vortexa's post-ingestion streaming estate, live-processing strategy, and model evaluation/replay interfaces downstream of **10+ AIS providers**. Scale teams and systems by *turning noisy data/model behaviour into auditable, observable, replayable production platforms*: clear standards, trusted monitoring, data/model versioning, safety mechanisms, and healthier delivery systems. Active UCL researcher and JTC'21 committee expert at the intersection of **AI and maritime analytics**, advocating standards that make models auditable, versioned, explainable, and adoptable.

work-experience

since 12/2020 **Vortexa Ltd, London, UK**

Engineering Manager / ML Tech Lead | Pod Lead

Manage Turingeries, a **five-person MLE/DS/DE pod** with daily SME/analyst partnership. Own team delivery and **live-processing strategy** for mission-critical post-ingestion streaming and ML inference systems downstream of **10+ AIS providers**, covering prediction trust, model evaluation/replay, monitoring, and product enrichment.

- **People & Talent**: Built the pod and helped grow the data-products department from **4 to 30+**; hired **5 senior DS/DEs**, evaluated **31 candidates**, mentored 4 graduates/interns, retained/progressed the full team, and on-boarded new joiners to PRs within a week.
- **Streaming Platform Strategy**: Introduced/scaled **Flink's** dataflow model across live pipelines, reducing cognitive load and maintenance effort while decoupling compute scaling from Kafka partition counts.
- **Trustworthy ML Operations**: Build the trust loop for destination/ETA predictions: **versioned models/data**, replay/evaluation interfaces, online monitoring, analyst feedback, controlled switch-over, and explainability work.
- **Reliability & Delivery Health**: Maintain monitoring, runbooks, rollback/fallback plans, and Jira alert workflows; keep **MTTR under 30 min** while using **DORA metrics** to expose bottlenecks, reduce toil, and protect team health.
- **Strategic Portfolio & Interfaces**: Turn ambiguous product/model friction into **evaluated workstreams**; align product, analysts, SMEs, frontend, and backend around ownership, prediction semantics, and product-ready interfaces.
- **Engineering Standardisation**: Scale delivery through Flink/Airflow archetypes, ADRs, docs close to code, tests-as-docs, dev containers, Backstage adoption, and DynamicIO data contracts/SLAs; cut a **3h DAG feedback loop to under 5 min**.
- **Architecture & Org Influence**: Instigated a weekly company architecture forum and influenced topology/re-org work around pod, analyst/SME, frontend, and product boundaries.

2016–2020 **DataReply, London, UK**

Data Scientist | ML Engineer

Mostly hands-on consultant across UBS, Vodafone, and CNHi; leadership was project-specific: lead DS/Scrum Master at CNHi, DS-platform contributor at Vodafone, and sole embedded consultant at UBS. Focused on productionising data science.

- **Vodafone Group**: Worked on Infinity, a GCP/**Kubeflow** DS platform, bringing hands-on DS workflow feedback into ML lifecycle support; built Red Agent, a mobile-network feature-engineering framework.
- **CNHi**: Lead Data Scientist/Scrum Master on predictive maintenance; managed/guided a DS team using **PySpark** over live vehicle sensory data with product-facing backlog discipline.
- **UBS**: Hands-on graph/process-mining analytics and observability work, building real-time pipelines with **Kafka**, **Elasticsearch**, and Python; later sole embedded DataReply consultant.

2010–2016 **Roles in UK Institutions**

Assoc. Lect. @GSM | Coding Teacher @DGC | TA @KCL|UCL

Taught Java, Python, data structures, systems architecture, semantic web, SQL, and AI at GSM, DGC, KCL, and UCL; developed clear technical communication habits during doctoral work.

education

- 2010–2014 **Ph.D. in Computer Science** [King's College London, UK](#)
Thesis: *Persuasion Dialogues & Opponent Modelling.*
Developed logical inference methods for large **Knowledge Graphs**, combining symbolic reasoning, **graph analytics**, and Bayesian techniques (e.g., MCMC). Research contributed to scalable AI systems grounded in formal semantics. Received **Best Poster Award** at IJCAI 2013 (413 submissions). Awarded **EPSRC PhD Scholarship**, **Outstanding TA Award** (student-nominated), and earned the **Graduate Certificate in Academic Practice**.
[Access Publications on Google Scholar](#)
- 2004–2010 **Diploma (BEng) in Computer Engineering** [University of Thessaly, GR](#)
Five-year polytechnic degree in Computer Science and Engineering with a **strong focus on mathematics** (10 courses), covering theoretical aspects, networking, and telecommunications. **Majored in Artificial Intelligence.**
- 2002–2004 **Military Officer: Second Lieutenant** [National Guard, CY](#)
Completed national service as a commissioned Second Lieutenant.

committees/affiliations

- since 10/2024 **University College London, Dept. of Information Studies** [Associate Researcher](#)
Promote research at the intersection of AI and maritime intelligence, with focus on trustworthy AI application, standardisation, and industry/academic collaboration.
- since 01/2021 **ISO - International Organisation for Standardisation** [Committee Expert Member for JTC'21](#)
Member of JTC'21 Working Group 3, developing AI standards aligned with EU policies and international norms.

talks/interviews

- 2023 **Bill Raymonds, Agile in Action** [Podcast Interview](#)
Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa, **Details:** [YouTube](#)
- 2022 **Open Data Science Conference** [Industry Conference Talk](#)
Dynamicio, a python library for abstracting your I/O, **Details:** [here](#)
- 2020 **iunera: Big Data Innovators** [Interview Blog Post](#)
The Agile approach in data science explained by an ML expert, **Details:** [here](#)
- 2020 **Big Data Warsaw** [Industry Conference Talk](#)
Monitoring & Analysing Communication and Trade Events as Graphs, **Details:** [here](#)
- 2018 **Connected Data London** [Panelist @ Industry Conference](#)
AI tribes and interconnections: what's graph got to do with it?, **Details:** [here](#)
- 2018 **Minds Mastering Machines** [Industry Conference Talk](#)
A talk on doing Data Science the Agile Way, **Details:** [here](#)

certificates & awards

- 07/2020 **AWS Certified Machine Learning - Specialty** [AWS](#)
Validation Number: D0R0WC7CN1VQ13SV
- 01/2019 **Process Mining** [Coursera Course Certificates](#)
License number: AFU2VTP6273V
- 10/2017 **Google Certified Professional - Data Engineer** [Google Cloud Certified](#)
Credential ID: 000000400
- 07/2017 **Advanced Elasticsearch: Data modelling** [Elastic](#)
Enrolment Id: 33442
- 06/2016 **Graph Analytics for Big Data** [Coursera Course Certificates](#)
License number: NZADYAQ7N8FW
- 05/2016 **Apache Spark Developer Certification** [O'Reilly Media](#)
License number: 1.x-0548
- 04/2016 **Neo4j Certified Professional** [Neo Technology, Creators of Neo4j](#)
License number: here